

FRANCESCO BANDERA

MSc Student in Music and Acoustic Engineering | francesco2.bandera@mail.polimi.it | github.com/Frabbandera | Italy

PROFILE

MSc student in Music and Acoustic Engineering at Politecnico di Milano with an Energy Engineering background and strong academic performance. Focused on acoustic simulation, vibroacoustics, vehicle acoustics, numerical modelling and machine-learning workflows for acoustic data. Experienced in FEM-based acoustic modelling, modal/FRF interpretation, RIR analysis, Python/MATLAB technical workflows, and validation-oriented engineering reporting. Available for master's thesis work from September/October 2026; expected graduation July 2027.

EDUCATION

Politecnico di Milano - MSc in Music and Acoustic Engineering | Cremona, Italy | 2024-present | Current weighted average: 29.67/30

- Relevant coursework: Vehicle Acoustics, Numerical Modeling and Simulation in Acoustics (30L), Vibration Analysis and Vibroacoustics (29), Room Acoustics (29), Fundamentals of Acoustics (30), Selected Topics in Music and Acoustic Engineering.

Politecnico di Milano - BSc in Energy Engineering | Milan, Italy | 2020-2024 | Final grade: 91/110

- Engineering foundations in theoretical/applied mechanics, solid mechanics, fluid mechanics, measurement and industrial instrumentation, analytical and numerical methods, technical physics, and mechanical design principles.

SELECTED TECHNICAL PROJECTS

Vehicle Cabin Acoustics and NVH Analysis

- Modelled the internal acoustic field of a realistic car cabin across 50–8000 Hz using COMSOL FEM for low-frequency modal and forced response analysis, ray tracing for higher-frequency acoustic behaviour, and passenger/scattering modelling.
- Investigated porous-trim sensitivity by varying thickness and flow resistivity, identifying mid-high frequency smoothing effects above 200 Hz and limited impact on low-frequency booming modes.
- Analysed structure-borne and airborne noise transmission through hammer-excitation FRFs, coherence analysis, vibrometer, body accelerometer and cabin microphone data, and engine-ramp order analysis.
- Interpreted spatial SPL non-uniformity, dominant engine order behaviour, and operational structure-borne transmission paths relevant to cabin acoustic comfort.

Numerical Acoustic Modelling and FEM Validation

- Developed a variational FEM solver for the Helmholtz equation in heterogeneous media, including a smooth quadratic PML to reduce artificial reflections at open-domain boundaries.
- Analysed wave reflection at material interfaces using impedance-mismatch theory and compared numerical behaviour with analytical expectations.
- Validated the implementation against analytical solutions, obtaining a convergence order of 2.17, consistent with theoretical expectations for P1 finite elements.

AI-Based Acoustic Signal Classification and Validation

- Contributed to a team-developed ML pipeline for 12-class instrument recognition, including metadata-based subset construction, audio indexing and segmentation, activity filtering, class balancing, and track-disjoint train/validation/test splits to prevent leakage.
- Analysed and reported the comparison between MFCC/spectral scikit-learn baselines, self-supervised MERT representations, and MLP classifier heads using validation-based model selection, per-class metrics, confusion matrices, and held-out test results.
- Supported representation and protocol ablations across Transformer layers, pooling strategies, learned layer aggregation, model sizes, partial fine-tuning, imbalance handling, and segment durations; best isolated-stem result reached 0.7290 test macro-F1.
- Extended the analysis to controlled synthetic polyphonic mixtures with one to four active instruments, evaluating robustness through macro/micro-F1, ROC-AUC, exact-match accuracy, and false-positive/false-negative interpretation.

TECHNICAL SKILLS

- Acoustic and Vibroacoustic Engineering:** vehicle cabin acoustics, NVH analysis, structure-borne and airborne noise, modal analysis, FRFs, order analysis, sound radiation, room acoustics, RIR/IR analysis, acoustic measurements, low-frequency modal behaviour.
- Simulation and Numerical Modelling:** COMSOL FEM and ray tracing, Helmholtz equation, PML boundaries, impedance mismatch, acoustic radiation prediction, Kirchhoff-Helmholtz integral, convergence analysis, analytical/numerical validation.
- Signal Processing and Data Analysis:** spectral analysis and descriptors, MFCCs, one-third-octave analysis, decay-curve interpretation, and error analysis.
- Machine Learning & AI:** supervised classification, transfer learning workflows, self-supervised audio representations, SVM/MLP classifiers, multi-label classification, ablation studies, validation-based model selection, per-class metrics and confusion matrices.
- Programming & Data Tools:** MATLAB, Python, C++, NumPy, pandas, scikit-learn, librosa, Git/GitHub, Jupyter/Colab. Academic project experience with COMSOL acoustic FEM/ray tracing; project exposure to PyTorch/Lightning and Hugging Face Transformers.
- Languages:** Italian native; English B2.

ADDITIONAL TRAINING

Room-Acoustic Measurements and Low-Frequency Treatment Design | Politecnico di Milano / Labirinti Acustici | 2026

- Completed hands-on training in impulse-response acquisition, ISO 3382 acoustic-parameter interpretation, CLIO/Earthworks measurement workflows, low-frequency modal analysis, and Helmholtz-type bass-trap tuning/validation.

ADDITIONAL EXPERIENCE AND LEADERSHIP

University Basketball Team - Player and Team President | 2020-2026

- Combined competitive sport with a demanding engineering workload; participated three times in the Italian National University Championship and won the tournament in May 2026.
- As 2025/2026 team president, coordinated facility search, cost forecasting, sponsor contacts, and communication with university bodies to keep the activity structured and financially sustainable.